AEO “Nazarbayev Intellectual Schools”

STICK BARCODE HERE

CENTRE NUMBER							CANDIDATE NUMBER				

4094148218

CHEMISTRY **Grade 12**

Paper 3 **May 2023**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
Qualitative Analysis Notes and Periodic Table
Ruler

12CHEM/03

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
You may receive fewer marks if you do not show your working or do not use appropriate units where required.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use	
1	
2	
Total	

This document consists of **9** printed pages and **3** blank pages.

For Examiner's Use

Read all the instructions for Question 1 carefully before starting the experiment.

1 In this experiment you will find the percentage purity of a sample of sodium sulfite, Na_2SO_3, by titration with acidified potassium manganate(VII), $KMnO_4$.

The ionic equation for this reaction is shown.

$$5SO_3^{2-} + 2MnO_4^{-} + 6H^{+} \rightarrow 5SO_4^{2-} + 2Mn^{2+} + 3H_2O$$

(a) Container A contains impure sodium sulfite, Na_2SO_3.

Weigh container A and its contents. Record this value.

Pour the impure sodium sulfite into a 250 cm^3 beaker.

Weigh empty container A. Record this value.

Calculate the mass of impure sodium sulfite transferred to the beaker.

Record these mass readings in an appropriate table.

[2]

(b) Add about 100 cm^3 of distilled water to the beaker. Stir the mixture with a stirring rod until all of the sodium sulfite has dissolved.

Transfer this solution into a 250 cm^3 graduated flask. Wash the beaker with small amounts of distilled water and transfer this water into the graduated flask.

Then fill the graduated flask up to the mark with distilled water.

Put a stopper on the graduated flask and thoroughly shake the mixture. Label this solution **B**.

Fill a burette with solution **B**. Record the initial reading of solution **B** in the burette.

Transfer 25.0 cm^3 of the 0.0200 mol dm^{-3} $KMnO_4(aq)$ into a 250 cm^3 conical flask using a pipette.

Use a measuring cylinder to add 15.0 cm^3 of 1.00 mol dm^{-3} sulfuric acid to the conical flask.

Titrate the 25.0 cm^3 of $KMnO_4(aq)$ against solution **B**. The end-point is the first appearance of a persistent colourless solution.

Record the final reading on the burette and determine the volume (titre) of solution **B** added.

Repeat until you obtain concordant results.

For Examiner's Use

(i) Record the results from the titrations in this table. You do not need to use all columns.

titration number						
initial (burette) reading / cm^3						
final (burette) reading / cm^3						
volume of solution B added / cm^3						

[7]

(ii) Calculate the mean volume of solution **B** added.

State which titration numbers you used to calculate this mean volume.

Give your answer to **two** decimal places.

titration numbers ....

mean volume .... cm^3 [2]

(c) **(i)** Calculate the amount, in moles, of $KMnO_4$ in 25.0 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$ solution.

.... mol [1]

(ii) Calculate the amount, in moles, of Na_2SO_3 in the mean volume of solution **B** added.

Use your answer to **(c)(i)** and the equation for the reaction.

.... mol [1]

(iii) Calculate the amount, in moles, of Na_2SO_3 in the 250 cm^3 of solution **B**.

.... mol [1]

(iv) Hence, calculate the mass of Na_2SO_3 in 250 cm^3 of solution **B.**

.... g [1]

For Examiner's Use

(d) **(i)** Calculate the percentage purity of the Na_2SO_3.

Show your working.

.... % [1]

(ii) Uncertainty of measurement of a burette is $\pm 0.05\,cm^3$.

Calculate the percentage uncertainty in the mean volume of **B** added during the titration.

Show your working.

.... % [2]

[Total: 19]

Read all the instructions for Question 2 carefully before starting the experiment.

For Examiner's Use

2 In this experiment you will determine the order of reaction with respect to sodium thiosulfate, $Na_2S_2O_3$, for the reaction with hydrochloric acid represented by the equation shown.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

(a) (i) Carry out the experiments and record your results in Table 2.1. All times recorded in whole numbers **and** in seconds.

Step 1. Draw the letter X on a piece of white paper.

Step 2. Use a large measuring cylinder to transfer 40.0 cm^3 of $Na_2S_2O_3(aq)$ into a 100 cm^3 beaker.

Step 3. Use a clean large measuring cylinder to transfer 10.0 cm^3 of $H_2O(l)$ into the beaker.

Step 4. Place the beaker over the piece of white paper with the letter X on it.

Step 5. Use a 10.0 cm^3 measuring cylinder to measure 5.0 cm^3 of $HCl(aq)$.

Step 6. Pour the 5.0 cm^3 of $HCl(aq)$ into the beaker **and** at the same time start a stop-watch.

Step 7. Immediately stir the mixture for 3 seconds.

Step 8. Record the time, to the nearest second, when the letter X is no longer visible when looking from above the beaker.

Step 9. Empty the mixture in the beaker labelled waste and thoroughly rinse the reaction beaker with water.

Step 10. Dry the beaker with a paper towel.

Repeat steps 2 to 10 with different volumes of $Na_2S_2O_3(aq)$ and $H_2O(l)$ as shown in Table 2.1.

Table 2.1

experiment number	**volume of $Na_2S_2O_3(aq)$ / cm^3**	**volume of $H_2O(l)$ / cm^3**	**volume of $HCl(aq)$ / cm^3**	**time / s**	**rate of reaction / s^{-1}**
1	40.0	10.0	5.0		
2	30.0	20.0	5.0		
3	25.0	25.0	5.0		
4	20.0	30.0	5.0		
5	15.0	35.0	5.0		
6	10.0	40.0	5.0		

[5]

For Examiner's Use

(ii) The rate of reaction is calculated using this expression.

$$\text{Rate of reaction} = \frac{1000}{\text{time}}$$

Calculate the rate of reaction for experiments 1 to 6.

Record your results in Table 2.1 to **two** significant figures.

[1]

(b) **(i)** Calculate the values of the log(volume of $Na_2S_2O_3$(aq)) and log(rate of reaction) and write them in Table 2.2.

Table 2.2

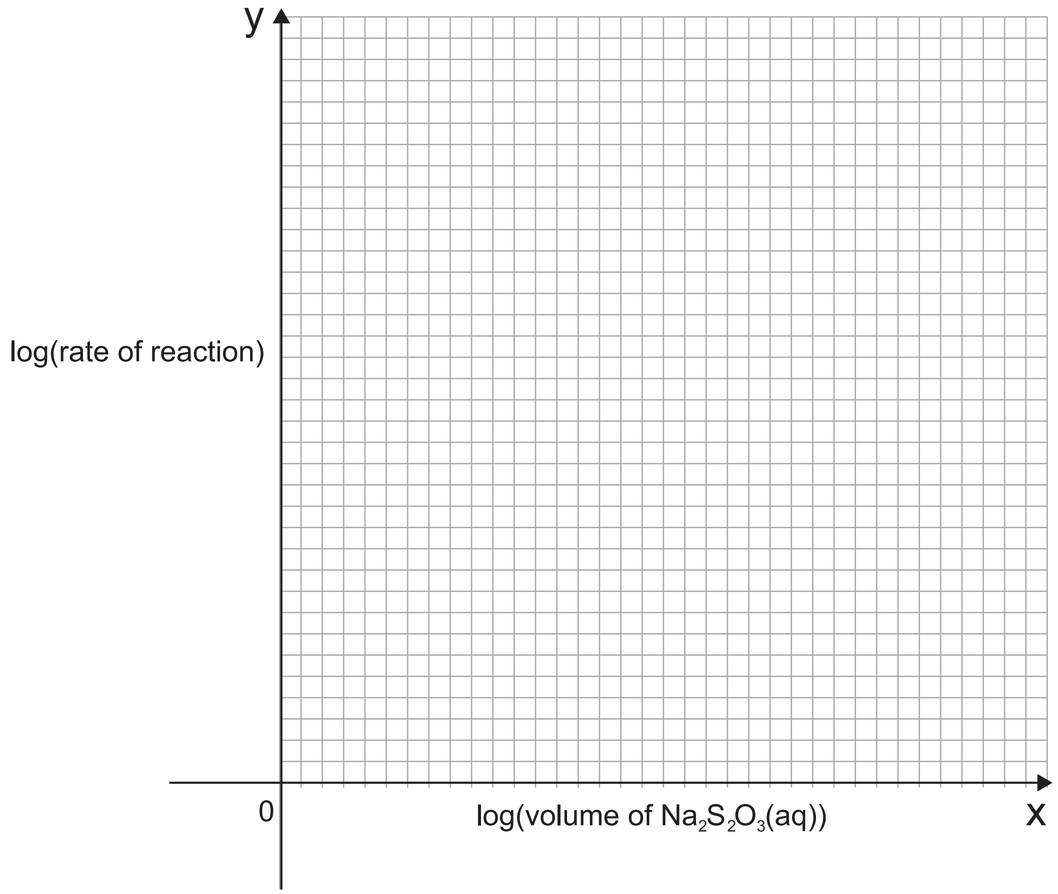
experiment number	**log(volume of $Na_2S_2O_3$(aq))**	**log(rate of reaction)**
1		
2		
3		
4		
5		
6		

[2]

For Examiner's Use

(ii) Plot log(rate of reaction) against the log(volume of $Na_2S_2O_3$)(aq)).

Draw a line of best fit.



[3]

(iii) The order of reaction with respect to $Na_2S_2O_3$ is the gradient of this graph.

Use your graph to suggest the order of reaction with respect to $Na_2S_2O_3$.

....

.... [1]

(iv) Write the rate equation for this reaction, when [HCl] is kept constant.

For Examiner's Use

[1]

(v) State the value of the **y** intercept.

Use your graph to determine the value of the **y** intercept. This is the value of log(rate of reaction) when log(volume of $Na_2S_2O_3$) is 0.

Value of log(rate of reaction) from the **y** intercept is .... [1]

(vi) The value of the y intercept is equal to log(rate constant) for this reaction.

Calculate the numerical value of the rate constant, *k*, and state its units.

numerical value .... unit .... [1]

For Examiner's Use

(c) You are provided with samples of two colourless solutions, **X** and **Z**.

These solutions are:

- zinc sulfate
- aluminium chloride.

Use information from the qualitative analysis booklet to plan and carry out different chemical tests to identify each of the solutions.

Each solution must be identified using **positive** chemical tests.

You can use:

- any piece of equipment found in a school laboratory;
- any chemicals found on the qualitative analysis booklet.

(i) Record your results and conclusions for solution **X** and **Z** in the table.

Specify which solution, **X** or **Z**, you have used in each test.

solution used	**chemical test**	**observation**	**conclusion**

[6]

[Total: 21]

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